Paper A v0.4 — The Physical (SU(3)) Yang–Mills Mass Gap via D_IV^5 Spectral Geometry. Companion to Paper B. v0.4 = v0.3 with F1056 folded in: (1) sentence one now states the TWO same-name guards (λ_1-on-D_IV^5 $\neq$ the YM Hamiltonian on $\mathbb{R}^4$; glueball $\neq$ proton) and the ONE residual (proving the reconstruction's edge-theory is literally interacting SU(3) YM), explanatory-framed; (2) the mass-gap operator is named precisely — the Bergman–Laplacian = Casimir of K=SO(5)×SO(2) (H_B, its positivity is the arrow) on H^2(D_IV^5), winding tower λ_k=k(k+n_C)=k(k+5), λ_0=0 (unwound vacuum), λ_1=C$_2$=6 (one winding), boundary conditions = bounded domain / reproducing-kernel-Shilov / discrete-spectrum-bounded-below, compactness IS the gap mechanism; (3) the residual is UNIFIED — 'Rehren-edge = literally-YM' (F1056) and 'interacting vs generalized-free' (v0.3 §6/§9) are ONE residual seen two ways, with the honest W4 stack named (Hua boundary values → Bisognano–Wichmann/Borchers modular geometry → Rehren holographic net → OS reconstruction → cluster decomposition). Explanatory claim: BST REPRODUCES the gap's scale on a bounded domain; the flat-$\mathbb{R}^4$ construction is the large named open, never a proof. Δ=C_2=6 SOLID; glueball 1720 @ 0.6% is the Clay object (not the proton 938). Count HOLDS 4.

Lyra (Claude Opus 4.8) — Casey Koons, PI; Grace (OS reduction + T2490/T2491 spectral casc

2026-08-19 Wednesday (date-verified)

The Physical Yang–Mills Mass Gap via D_IV^5 Spectral Geometry

Paper A of the two-paper package (companion: Paper B, substrate uniqueness). v0.4 — F1056 folded: operator named precisely, two same-name guards and one residual up front, explanatory framing.

0. Two guards and one residual (read this first)

Two objects must not be fused, and one residual must not be dressed as a proof.

- Guard 1 (space). The gap we derive is the spectral gap of the Bergman–Laplacian on the bounded domain D_IV^5 — a *different operator on a different space* from the Yang–Mills Hamiltonian on flat $\mathbb{R}^4$. $\lambda_1 = C_2 = 6$ on our domain is not, by assertion, the $\mathbb{R}^4$ mass gap; the two are connected only through a bridge that must be exhibited, never fused (the standing K932 guard: exhibit the bridge, don't assert a shared eigenvalue).
- Guard 2 (object). The pure-Yang–Mills (Clay) gap is the glueball (1720 MeV, 0.6% vs lattice), not the proton (938 MeV, which is the *full-QCD*, matter-included gap — a different object). We do not lend the proton's

four-decimal precision to the Yang–Mills gap (K1705).

- The one residual. The honest open problem is a single question: is the theory that the reconstruction machinery (Rehren holography → Osterwalder–Schrader) places on the flat 4D boundary literally interacting SU(3) Yang–Mills, or a gapped generalized-free cousin that also satisfies the axioms? This is the large named open — carried as such, never a proof.

With those stated, the paper's claim is explanatory, not triumphal: BST reproduces the scale and channel-structure of the Yang–Mills mass gap on a bounded domain. The gap value $\Delta = C_2 = 6$ is forced by representation theory; its pure-glue reading matches the glueball at 0.6%; the flat-$\mathbb{R}^4$ construction is carried as the large named open. Nothing here is "we solved Yang–Mills."

Abstract

We study the Yang–Mills mass gap for the physical gauge group SU(3) — color — derived (not assumed) from the substrate D_IV^5 in the companion Paper B. The mass-gap operator is named precisely (Section 2): the Bergman–Laplacian, the Casimir of the maximal compact K = SO(5)×SO(2), acting on the Bergman–Hardy space H^2(D_IV^5), with winding spectrum λ_k = k(k+n_C) = k(k+5); the unwound vacuum sits at $\lambda_0 = 0$ and one winding at $\lambda_1 = C_2 = 6$, so the gap is the exact integer $\Delta = C_2 = 6$. Because the domain is bounded, the spectrum is discrete and bounded below — *compactness is the gap mechanism* (decompactify and the gap slides to zero). The construction does not build 4D Yang–Mills from scratch on flat $\mathbb{R}^4$; instead the D_IV^5 spectral theory directly supplies Osterwalder–Schrader reconstruction data (a rigorous Hilbert space, a gapped Hamiltonian, reflection positivity, 4D conformal/Poincaré covariance, clustering) on the domain's 4D conformal boundary (SO(4,2) ⊂ SO(5,2)), so the existence of a rigorous gapped 4D theory follows from classical harmonic analysis on a bounded symmetric domain. The problem then reduces to a single residual (Section 6, Section 9): is this gapped edge theory genuinely interacting Yang–Mills, or a (also-gapped, also-axiom-satisfying) generalized-free theory — equivalently, is the Rehren-reconstructed boundary theory *literally* SU(3) YM? In v0.4 we report that the decisive probe — the cross-channel glueball spectrum — has been computed and materially advances the interacting reading: the four ground J^PC channels follow a linear conformal-energy ladder off one verified number (the genus of D_IV^5), with one blind leg, 2^{++} = g/n_C, and the other channels consistent within current lattice error. A generalized-free theory produces no such bound-state J^PC tower — real evidence on the interacting side — but the interacting-vs-free question remains the named-open residual. We do not claim the Clay problem is solved. The honest statement is a decisive *sharpening* to one residual, with the flat-$\mathbb{R}^4$ identification carried as the large named open.

1. Scope: the physical theory, not "all G"

The Clay problem asks for a mass gap for *every* compact simple gauge group. BST addresses the physical theory: G = SU(3), color. The companion Paper B proves

D_IV^5 is the unique substrate and that SU(3) is derived, not chosen. So "all G" is not BST's claim. This paper makes claims only about the physical theory and is explicit about that scope (Section 9).

2. The mass-gap operator, named precisely, and its value Δ = C_2 = 6 (SOLID)

The operator. The mass-gap operator is the Bergman–Laplacian — equivalently the Casimir H_B of the maximal compact subgroup K = SO(5)×SO(2), whose positive spectrum *is* the arrow of time (the heat-semigroup generator) — acting on the Bergman–Hardy space H^2(D_IV^5), the holomorphic L^2 functions on the bounded domain (the glueball/winding wavefunctions).

- Spectrum (winding tower). λ_k = k(k+n_C) = k(k+5) = {0, 6, 14, 24, …} (the scalar-Laplacian tower on the compact dual Q^5 = SO(7)/[SO(5)×SO(2)]; k = winding number). The unwound vacuum sits at $\lambda_0 = 0$; one winding at $\lambda_1 = 1\cdot(1+5) = C_2 = 6$. The gap is $\Delta = \lambda_1 - \lambda_0 = C_2 = 6$ — the SO(7) vector Casimir, an exact integer from representation theory.
- Boundary conditions. The domain is bounded; H^2(D_IV^5) is defined by square-integrability against the invariant Bergman measure, with the reproducing-kernel (Cauchy–Szegő / Shilov-boundary) structure fixing boundary behavior. The spectrum is discrete and bounded below — and *this compactness is the gap mechanism*: decompactification ($Q^5 \to \mathbb{R}^4$, $R \to \infty$) sends $\lambda_1(R) = 6/R^2 \to 0$, killing the gap. The gap exists because the domain is bounded.

Two structural facts about the value:

- C_2 = 2·N_c (general-rank closed form, T2491). The (1,1)-type Casimir of SO(n) is 2(n−2) = 2·N_c; at the substrate's rank it equals 6. *(Value-recurrence note, Cal #335: C_2 = 6 also appears as N_c(N_c+1)/rank and as the k=1 rung of {k(k+5)}. These coincide via the rank-2 cascade — value-recurrences of one quantity, not independent derivations of C_2.)*
- Which physical gap this is (K1705/K1706 — Guard 2). The gap rung is physical, but the reading depends on matter content. The proton is the first *full-QCD* rung: m_p = C_2·π^5·m_e = 938.25 MeV (0.002%). The Clay object is the pure-glue glueball: the 0^{++} sits on the absolute seat ladder at seat c_2 = C_2 + n_C = 11 → 1720 MeV (T1788, 0.6% vs lattice 1710±50). *Cite the glueball at 0.6% as the Yang–Mills number; the proton (938, full QCD) is a different object.*

Tier: SOLID (rep theory + the named substrate-Hamiltonian identification). *This is a domain result — Guard 1: it is the gap of the Bergman–Laplacian on D_IV^5, not asserted to be the $\mathbb{R}^4$ YM Hamiltonian's gap.*

3. W4 dissolved: the gauge group is derived (Paper B)

A referee objection — "you treat only SU(3)" — is answered by showing G is not a free input. Paper B establishes D_IV^5 as the unique irreducible Hermitian symmetric domain meeting prior, dimension-innocent criteria, forcing $\dim\_C = 5$ and $N\_c = 3$ (the dimension-free output of the proved T1829). $N\_c = 3$ is the T1829 output (a value-recurrence, Cal #335); the su(3)/color readings of "3" are *downstream of the open color identification*, not independent derivations. So SU(3) is substrate-derived. The whole dynamical content cascades from one physical input — three colors fix rank = 2 ($N\_c = \text{rank}^2-1$), and rank generates {n_C, C_2, g} (T2491). The "all G" wall is dissolved.

4. W1 folded: the D_IV^5 spectral theory supplies the OS data on the 4D conformal boundary

BST does not construct 4D YM from scratch on $\mathbb{R}^4$. The OS reconstruction theorem builds a QFT from a data set, and the D_IV^5 spectral theory carries it — on the domain's 4D conformal boundary, which exists because D_IV^5 carries the 4D conformal group SO(4,2) ⊂ SO(5,2) (the same edge where light and electromagnetism live):

OS datum	D_IV^5 realization	tier
rigorous Hilbert space	the Hardy space $H^2(D\_IV^5)$	SOLID
Hamiltonian with a gap	the Bergman–Laplacian (Casimir of K), gap = $C\_2 = 6$	SOLID (Section 2)
reflection positivity	tube-type (Cayley → T(Ω)); Θ = cone involution; OS-RP (free level) = Cauchy–Szegő RKHS-positivity	SOLID (free level)
4D conformal/Poincaré	SO(4,2) ⊂ SO(5,2), UV-conformal/IR-gapped	SOLID-structural
clustering / unique vacuum	from the spectral gap $\Delta = C\_2 = 6 > 0$	SOLID

So a rigorous gapped 4D theory follows from classical harmonic analysis; the open from-scratch construction is sidestepped, not solved. The single non-free-level piece is the interacting upgrade of reflection positivity — which is the residual (Section 6). Tier: SOLID at free level; interacting-RP = the residual.

5. Net-compatibility via Bisognano–Wichmann (SOLID-CONDITIONAL)

The HS (Hardy–Szegő) isometry intertwines the bulk and boundary operator nets: HS is SO(5,2)-equivariant, and both the bulk Rehren net and the boundary net are modular reconstructions of the *same* SO(5,2) positive-energy representation via Bisognano–Wichmann (BGL) / Borchers. So locality/causality transfer across HS. Tier: SOLID-CONDITIONAL on the BGL hypothesis-checks. *(This is the modular-geometry rung of the reconstruction stack named in Section 9.)*

6. The one residual: interacting Yang–Mills, or generalized-free? (= "is the Rehren edge literally YM?")

The prize is a single question, and F1056 and v0.3 name it two ways that are one residual:

Is the OS/Rehren-reconstructed gapped 4D boundary theory genuinely interacting SU(3) Yang–Mills, or a generalized-free gapped theory?

The two namings coincide: - v0.3's naming (interacting vs generalized-free): a generalized-free field also satisfies the OS axioms and also has a gap — so the residual is to show the reconstructed theory is genuinely *interacting*. - F1056's naming (Rehren-edge = literally-YM): Rehren's algebraic holography is a rigorous bulk↔boundary net correspondence, but its *dual is not automatically the physical CFT* — so the residual is to show the Rehren edge theory is *literally* SU(3) YM, not a cousin.

These are the same open piece. "Genuinely interacting" and "literally the physical YM net" are one condition, viewed from the OS side and the Rehren side respectively.

A trap we avoid. "Non-commutative ==> interacting" is false — a free field also has a non-commutative CCR algebra. Interaction means nonzero connected n-point functions ($n \geq 3$). The YM self-interaction is the vertex [A,A], nonzero precisely because the gauge algebra is non-abelian. So

interacting <=> the gauge algebra is genuinely non-abelian su(3) <=> the cross-channel glueball spectrum matches SU(3) (Section 7).

Two pieces of evidence toward the interacting side: - The bulk-color algebra closes as non-abelian su(3) (bilinear Schwinger realization; Elie 4301). The substrate identification (#418) is in progress. - The spectrum is non-additive and computed (Section 7). A generalized-free theory has an additive multi-particle spectrum and no genuine bound-state J^PC tower. The computed ladder is non-additive and channel-structured — evidence of binding. This is *evidence toward* interacting, not a closure of it.

7. The cross-channel glueball spectrum — named-open, materially advanced

In v0.1 this section was *named-open at structural tier*. Since v0.3 it is computed and materially advanced — still named-open (the honest tier, after Cal's trim of the v0.2 over-claim "derived / landed interacting").

7.1 Why there are exactly four channels (operator-algebraic closure)

The gauge field F is a rank-2 antisymmetric tensor operator on $H^2(D\_IV^5)$. Its bilinears F⊗F decompose into exactly four irreducible J^PC components, each of which must have eigenstates for the substrate's operator algebra to close:

channel	operator	what it commits
0^{++}	$Tr(F^2)$	energy density (the action)
0^{-+}	$Tr(F\ \tilde{F})$	topology (Pontryagin density)
2^{++}	Tr(F^$\mu\rho$ F^ν_ρ) traceless	curvature (the stress tensor)
1^{+-}	Tr(F[D,F])	derivative structure

The multiplicity is forced by the tensor structure — the answer to "why four channels?" Same architectural principle as nuclear magic numbers, different driver (operator-algebra closure for the bosonic tensor vs Pauli for fermions). The oddballs (0^{+-}, 1^{-+}, 2^{+-}) are forbidden at two gluons — clean unmixable channels.

7.2 The masses: one verified number sets the ladder

The glueball mass is the eigenvalue of the linear conformal Hamiltonian (the SO(2) dilatation) on the holomorphic discrete series on $H^2(D\_IV^5)$, diagonal in the K-type basis by Schur:

m ∝ E = λ_0 + (energy step), λ_0 = genus(D_IV^5) = n_C = 5 (verified from the multiplicity formula p = (r−1)a + b + 2 = 5).

Energy step = SO(5) harmonic degree = spin J; the parity-odd (Hodge-dual) sector carries the half-canonical twist n_C/2. With one dimensionful anchor (seat = π^5·m_e = 156.4 MeV; m(0^{++}) = c_2·seat = 1720 MeV):

channel	step	ratio	substrate form	BST (MeV)	lattice (MeV)	dev	tier
0^{++}	0	1	—	1720	1730	−0.6%	anchor
2^{++}	J = 2	7/5	g/n_C	2408	2400	+0.3%	BLIND (genus + spin)

channel	step	ratio	substrate form	BST (MeV)	lattice (MeV)	dev	tier
0^{-+}	n_C/2	3/2	N_c/rank	2580	2590	−0.4%	consistent (twist value-checked)
1^{+-}	1 + n_C/2	17/10	—	2924	2940	−0.5%	consistent (twist value-checked)

Only the 2^{++} = g/n_C leg is fully blind (genus + spin only, nothing read from the data, lands because g = n_C + rank). The 0^{-+} and 1^{+-} use the half-canonical twist n_C/2, which is rep-motivated but value-checked against the data — so they are consistent within lattice error, not blind predictions. (Quantization note: the *ratios* live on the rung ladder λ_0 = n_C; the absolute scale comes only from anchoring 0^{++} at seat 11.)

7.3 The radial direction is linear, by Schur

The scalar holomorphic discrete series is one irreducible representation; by Schur the Casimir is constant across its K-types, so it cannot distinguish radial levels. The first radial scalar excitation 0^{++*} = (1,1) sits at the linear conformal energy E = λ_0 + 2 = 7, degenerate with 2^{++} (2408 MeV; lattice ~2670, within quenched excited-glueball error). Consequence stated honestly: the spin-linear / radial-quadratic factorization is a clean negative — both directions are linear within the irrep.

7.4 Experimental and decay-side notes

The experimental scalar-glueball candidate $f_0(1710) \approx 1704$ MeV is consistent with BST's 0^{++} at seat 11 (1720 MeV) — the expected lightest-glueball match (not a unique identification; the scalar sector mixes with $q\bar{q}$). The 0^{-+} coupling f_G is the substrate-computable Bergman mode norm (a Gindikin-Gamma ratio; 0^{++} kernel = 60 = C_2·n_C·rank); with the established χ_top^{1/4} = 180 MeV the WV identity gives f_G(0^{-+}) ≈ 12.6 MeV. Decay-side: mixing (glueball ↔ $q\bar{q}$ overlap on the shared Hardy space), not a “dump.”

7.5 Honest disposition of Section 7

The cross-channel spectrum is computed (four channels; one blind leg 2^{++} = g/n_C; the others consistent within lattice error; multiplicity explained; radial direction settled by Schur; lightest state consistent with $f_0(1710)$). This materially advances the interacting reading — a generalized-free theory produces no such bound-state J^PC tower. It does not close the interacting-vs-free residual (Section

6). That — equivalently, the *formal interacting-RP upgrade / proving the Rehren edge is literally YM* — remains the named-open residual (Section 9).

8. The so(7) unification (LEAD-STRENGTHENED)

Color and the Yang–Mills spectrum live in one algebra: su(3) $\subset g_2 \subset$ so(7), and so(7) is the compact dual isometry of Q^5 — the same so(7) whose Casimir spectrum is the glueball tower of Section 2; g = 7 is the so(7) vector = g_2 fundamental = $3 \oplus \bar{3} \oplus 1$. Color is not a geometric isometry of the noncompact domain (su(3) not-subset-of so(5,2)) but is a geometric subalgebra of the compact-dual so(7); on the domain side it is the operator algebra on H^2. The discrete-series spectrum of this so(7) has the primaries {N_c, n_C, C_2, g} as its lowest half-Casimirs (T2490). Tier: LEAD-STRENGTHENED — the algebraic picture is whole; the bilinear-Schwinger realization on H^2 is in progress (#418).

9. Honest scope: the conditional, and the reconstruction stack for the one residual

We do not claim to have proved the Yang–Mills mass gap. The honest statement:

Given the reconstruction stack executes to rigor — including the interacting upgrade that identifies the edge theory as literally SU(3) YM — the physical Yang–Mills theory exists as a rigorous 4D QFT with mass gap Δ = C_2 = 6, whose pure-glue scale is the glueball at 1720 MeV.

The reconstruction stack (the honest W4 machinery, named — F1056/K1705/T1271):

Hua-1963 boundary values → Bisognano–Wichmann-1975 / Borchers-2000 modular geometry → Rehren-2000 holographic net → Osterwalder–Schrader reconstruction → cluster decomposition.

Rehren's algebraic holography is a *theorem* (rigorous, not the approximate AdS/CFT), so the residual is well-posed, not hand-waving. But it has two named, unfinished parts: (1) run the correspondence on D_IV^5 (build the bulk net and obtain the 4D boundary net); (2) identify the reconstructed boundary theory with the physical SU(3) YM — the known subtlety that a Rehren/OS dual is not *automatically* the physical CFT. Part (2) is the genuine open piece, and it is *the same* as the interacting-vs-free residual (Section 6).

What is SOLID independent of the conditions: the gap *value* C_2 = 6 (Section 2); the substrate-derivation of SU(3) (Paper B); the OS *data* (Section 4); the so(7) unification (Section 8); and the glueball scale 1720 MeV at 0.6% (the Clay object). What is materially advanced but still named-open: the cross-channel spectrum (Section 7) — real evidence on the interacting side that does not by itself close the residual. The single residual is the interacting-upgrade / literally-YM identification. This is a sharpening of the Clay problem to one decidable formal step — *not* a claim of its solution.

Explanatory summary (Guard 1 + Guard 2 held). BST *reproduces the scale and channel-structure of the Yang–Mills mass gap on a bounded domain*: the gap $\Delta = C_2 = 6$ is forced by representation theory (Guard 1: a domain operator, not the $\mathbb{R}^4$ Hamiltonian), its pure-glue reading matches the glueball at 0.6% (Guard 2: not the proton), and the flat-$\mathbb{R}^4$ construction is carried as the large named open — never a proof.

10. Open items and falsifiers

- The one residual (Section 6/9): the interacting-upgrade / proving the Rehren-OS edge theory is literally SU(3) YM.
- #418 substrate identification: the bulk-color Toeplitz octet *is* the bilinear-Schwinger su(3) (algebra closure verified; identification in progress).
- Blind-leg audit promotion: confirm λ_0 = genus = n_C is the Bergman lowest weight and canonical-bundle weight = genus (Elie toys A+B) — would promote the $0^{-+}/1^{+-}$ legs from value-checked to blind.
- D_A dynamical-gap check (Elie, Round 12): does the fluctuated Dirac D_A^2 carry $\lambda_1 = C_2 = 6$ dynamically (gauging preserves the winding-energy gap)? Report "the D_IV^5 gap," target the glueball, not the proton.
- Falsifier: the spectrum is a forward prediction — 0^{++*} degenerate with 2^{++} at 2408 MeV is testable; if the cross-channel ratios fail, the interacting reading fails (the gap value C_2 = 6 and Paper B survive).

Count HOLDS 4 of 26. SU(3) scope; glueball masses are predictions, not SM parameter reductions. Paper A v0.4: F1056 folded — the mass-gap operator is named precisely (Bergman–Laplacian = Casimir of K on H^2(D_IV^5), λ_k=k(k+5), λ_1=C_2=6, compactness = the gap mechanism); sentence one carries the two same-name guards and the one residual; the residual is unified (Rehren-edge = literally-YM ≡ interacting-vs-free). The claim is explanatory — BST reproduces the gap's scale on a bounded domain — with the flat-$\mathbb{R}^4$ construction as the large named open. Gap value SOLID; Clay object = glueball 1720 @ 0.6% (not proton). NOT "spectrum closed," NOT "YM proved." INTERNAL. Nothing pushed; CP existence-only.

Draft v0.4 (F1056 fold of v0.3). Spine: Grace's OS reduction + net-compat + T2490/T2491 + L2 + C_2=6 primary-Casimir pin; Lyra's named operator (F1056) + linear-conformal-Hamiltonian ladder + Schur radial settlement + residual unification; Elie's Schwinger closure + proton-gap-rung + D_A blind spectrum; the so(7) unification; Paper B (gauge-group derivation); operator-algebraic-closure multiplicity (K498); Keeper K1705/K1706 (gap=6, glueball not proton). Companion to Paper B v0.6.

— Lyra, Wed 2026-08-19 (date-verified). Paper A v0.4, F1056 folded, explanatory framing per Casey Round 12.